Practical No. 4 Implement a Naive Bayes Classifier and Evaluate the Performance

Submitted By: Shantanu Yadav
Roll No.: 75

In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = pd.read_csv('plant_growth_data.csv')
df
```

Out[2]:

	Soil_Type	Sunlight_Hours	Water_Frequency	Fertilizer_Type	Temperature	Hu
0	loam	5.192294	bi-weekly	chemical	31.719602	61.
1	sandy	4.033133	weekly	organic	28.919484	52.
2	loam	8.892769	bi-weekly	none	23.179059	44.
3	loam	8.241144	bi-weekly	none	18.465886	46.
4	sandy	8.374043	bi-weekly	organic	18.128741	63.
...	...	...	...	...	...	
188	sandy	5.652000	daily	none	28.000000	70.
189	clay	7.528000	weekly	chemical	30.500000	60.
190	loam	4.934000	bi-weekly	none	24.500000	61.
191	sandy	8.273000	daily	organic	27.900000	69.
192	clay	6.732000	weekly	none	21.700000	56.

193 rows × 7 columns

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 193 entries, 0 to 192
Data columns (total 7 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Soil_Type         193 non-null    object 
 1   Sunlight_Hours    193 non-null    float64
 2   Water_Frequency   193 non-null    object 
 3   Fertilizer_Type   193 non-null    object 
 4   Temperature       193 non-null    float64
 5   Humidity          193 non-null    float64
 6   Growth_Milestone  193 non-null    int64  
dtypes: float64(3), int64(1), object(3)
memory usage: 10.7+ KB
```

In [4]:
```
df.shape
```

Out[4]:
```
(193, 7)
```

In [5]:
```
df.describe()
```

Out[5]:

	Sunlight_Hours	Temperature	Humidity	Growth_Milestone
count	193.000000	193.000000	193.000000	193.000000
mean	6.826484	25.076087	58.098927	0.497409
std	1.599509	5.354170	12.631799	0.501294
min	4.033133	15.200000	30.567682	0.000000
25%	5.477000	20.637095	49.300000	0.000000
50%	6.833290	25.912336	59.182806	0.000000
75%	8.241144	29.757938	69.100000	1.000000
max	9.913903	34.810103	79.648240	1.000000

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
Soil_Type           0
Sunlight_Hours      0
Water_Frequency     0
Fertilizer_Type     0
Temperature         0
Humidity            0
Growth_Milestone    0
dtype: int64
```

In [7]:
```
df.columns
```

Out[7]:
```
Index(['Soil_Type', 'Sunlight_Hours', 'Water_Frequency', 'Fertilizer_Typ
e',
       'Temperature', 'Humidity', 'Growth_Milestone'],
      dtype='object')
```

In [8]:
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

df['Soil_Type'] = le.fit_transform(df['Soil_Type'])
df['Water_Frequency'] = le.fit_transform(df['Water_Frequency'])
df['Fertilizer_Type'] = le.fit_transform(df['Fertilizer_Type'])
```

In [9]:
```
X = df.drop(columns = ['Growth_Milestone'],axis = 1)
y = df['Growth_Milestone']
```

In [12]:
```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, confusion_matrix, classificat
from sklearn.neighbors import KNeighborsClassifier
```

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random

scaler = StandardScaler()
x_train_scaled = scaler.fit_transform(X_train)
x_test_scaled = scaler.transform(X_test)
```

In [13]:
```
models = {
    "KNN":KNeighborsClassifier(),
    "Naive Bayes":GaussianNB(),
}
```

In [14]:
```
results = []
```

In [15]:
```
for name,model in models.items():
    model.fit(x_train_scaled,y_train)
    y_pred = model.predict(x_test_scaled)
    acc = accuracy_score(y_test,y_pred)
    f1 = f1_score(y_test,y_pred)
    cm = confusion_matrix(y_test, y_pred)
    results.append({
        'model':name,
        'accuracy':round(acc,4),
        'f1 score':round(f1,4),
'confusion matrix': cm
    })
```

In [16]:
```
results
```

Out[16]:
```
[{'model': 'KNN',
  'accuracy': 0.4872,
  'f1 score': 0.4737,
  'confusion matrix': array([[10, 10],
         [10,  9]])},
 {'model': 'Naive Bayes',
  'accuracy': 0.4872,
  'f1 score': 0.5,
  'confusion matrix': array([[ 9, 11],
         [ 9, 10]])}]
```